Robust 3D Vessel Aneurysm Classification with Voxel and Point Representations

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Abstract

This project studies binary classification of 3D intracranial vessel segments as normal vessel or aneurysm using the VesselMNIST3D benchmark. We compare two representations of the same 3D input: a voxel-based 3D convolutional neural network and a point-based PointNet pipeline built from voxel-to-point conversion. Relative to the briefing-stage system, we improve robust training with a ramped augmentation schedule, clean/corrupted batch mixing, threshold calibration, and a richer evaluation suite. On the clean test set, the robust voxel model reaches 0.9356 AUROC and 0.9215 accuracy; with test-time augmentation (TTA), clean AUROC further increases to 0.9493 with similar accuracy (0.9241). The robust point model (xyz+I, 512 points) achieves 0.8657 AUROC and 0.8691 accuracy. Overall, the voxel branch is the strongest robust baseline under our controlled shift benchmark, while the point branch remains a meaningful geometric alternative that is more sensitive to design choices.

Attestation of Teamwork

All group members contributed to problem framing, literature review, implementation planning, and experimental design. Jenny primarily handled dataset loading, preprocessing, and the voxel-based pipeline implementation. Jasmine primarily worked on point-cloud conversion, PointNet experiments, and feature ablation settings. Yuye primarily handled benchmark pipeline standardization, threshold calibration, result analysis, experiment design, and report drafting.

1 Introduction

Intracranial aneurysm screening from 3D vascular data is a clinically meaningful classification task because missed abnormal vessel segments can delay diagnosis, while false positives increase review burden. Deep learning is suitable because the target signal is spatial and structural: vessel morphology,

branching geometry, and local aneurysm shape are encoded in 3D rather than in isolated handcrafted features. This project investigates whether the task is better handled directly in voxel space or after converting the same data into a point-cloud representation.

Voxel-based 3D CNNs are the standard baseline for volumetric medical imaging (e.g., 3D U-Net and V-Net) [4,5]. However, vessels are thin and geometric structures, which motivates testing a sparse point-cloud representation processed by PointNet [2] as an alternative input. Our goal is to compare these two pipelines under controlled distribution shifts (rotation and additive corruption) and identify which representation is more robust.

Our earlier briefing established that both pipelines were operational but that robust training still needed work. The final project therefore focuses on turning robustness into a first-class part of the comparison. The codebase was improved with a milder robust-training curriculum, mixed clean/corrupted batches, threshold calibration, and more complete metric logging. These changes let us revisit the original question under a better-balanced training and evaluation setup. The final results show that the voxel branch now provides a strong robust baseline, while the point branch remains informative but more sensitive to design choices.

2 Preliminaries and Problem Formulation

The dataset used in this project is VesselMNIST3D, a binary 3D medical-image benchmark distributed through MedMNIST [1, 3]. Each example is a single-channel volume

$$x \in \mathbb{R}^{1\times 28\times 28\times 28},$$

with label

$$y \in \{0, 1\},$$

where class 0 denotes normal vessel and class 1 denotes aneurysm [3]. The official split contains 1335 training samples, 191 validation samples, and 382 test samples [3].

The goal is to train a classifier

$$f_\theta(x) \in [0, 1]$$

that predicts aneurysm probability from a 3D input volume. Because the dataset is imbalanced and the operating point can change under weighted loss, AUROC is used as the primary validation metric and accuracy as a secondary metric. In the finalized code, we additionally log balanced accuracy, precision, recall, and F_1 score, and we calibrate the classification threshold on the validation set instead of always fixing it at 0.5.

A central design question is the representation of 3D structure. The voxel formulation preserves the original dense grid and is naturally aligned with 3D convolution. The point formulation first extracts salient voxels, converts them into a fixed-size point set, and then applies a PointNet-style network [2]. This comparison tests whether an explicit geometric representation can match a volumetric baseline on both clean and shifted data. We also consider robustness under controlled perturbations. Let

$$\tilde{x} = T(x)$$

denote a transformed input where T applies random rotation and additive corruption (Gaussian noise in our controlled stress test), or both. The robustness objective is to evaluate whether models trained on clean or augmented data maintain performance under these shifts.

3 Design

The system has a shared input stage followed by two downstream branches. Each sample is loaded in its native $1 \times 28 \times 28 \times 28$ format, and the same corruption utilities are used across experiments so that voxel and point models are evaluated under identical rotation and noise conditions. The point-cloud input is generated from the already-corrupted volume so that both branches observe equivalent perturbations.

The voxel branch is a compact 3D convolutional neural network composed of stacked Conv3D–BatchNorm–ReLU blocks with max-pooling, followed by adaptive pooling and fully connected classification layers. This branch serves as the dense volumetric baseline. The point branch converts

a volume into a fixed-size point cloud by thresholding salient voxels, falling back to top-k candidates if needed, normalizing coordinates, and optionally appending intensity as an additional feature. The resulting point set is processed by a PointNet-style classifier with shared multilayer perceptrons, symmetric max pooling, and a classification head; the main configuration uses 512 points with (x, y, z, I) features.

The finalized design also includes two robustness components. First, train-time augmentation applies random 3D rotations and additive Gaussian noise. Compared with the briefing version, the final code uses a milder schedule and a clean/corrupted batch mix so that robust training no longer destroys clean performance. Second, TTA averages predictions over multiple perturbed views at inference time. This makes robustness part of both training and evaluation.

4 Methodology

The project is implemented in Python using PyTorch. Data are acquired through the MedMNIST API, which provides the official VesselMNIST3D split and labels [3]. Both pipelines use mini-batch training with Adam, weight decay, class-weighted cross-entropy, mixed precision when available, and early stopping based on validation AUROC. The point pipeline operates on fixed-size point sets produced by voxel-to-point conversion.

Several improvements were added after the briefing. First, robust training uses a ramped augmentation schedule instead of applying strong corruption immediately. Second, each robust batch mixes clean and corrupted examples to stabilize optimization and preserve clean-set behavior. Third, we apply validation-based threshold calibration so that thresholded metrics are less distorted by class imbalance. Finally, the notebook exports a more complete result package (additional metrics, thresholds, summaries, and robust-vs-TTA comparisons), improving the reliability and interpretability of the final evaluation.

5 Numerical Experiments

We evaluate four main models: `voxel_clean`, `voxel_robust`, `point_clean_xyzI_512`, and `point_robust_xyzI_512`. The point configuration (xyz+I, 512 points) is selected based on point-construction ablations and then used as the final point baseline for the main voxel–point comparison. The benchmark includes the clean test set and two controlled shifts: random 3D rotation (maximum angle $0°$, $15°$, $30°$, $45°$) and an additive corruption stress test (Gaussian noise with standard deviation 0, 0.03, 0.06, 0.10). While real clinical variability is often driven by scanner/protocol and preprocessing differences, these controlled shifts provide a reproducible way to compare robustness trends across models.

5.1 Including Figures

Figure 1 shows AUROC versus severity for the four main models under rotation and additive corruption. Robust training reduces performance degradation under shifts, with the clearest gains in the voxel branch. Figure 2 summarizes robustness using average AUROC drop, providing an actionable criterion for model selection: the robust voxel model has the smallest average drop, while robust point training substantially reduces the large drop observed for the clean point model.

5.2 Including Tables

Table 1: Clean-set results for the main final models.

Model	AUROC	Accuracy
Voxel clean	0.8935	0.8770
Voxel robust	0.9356	0.9215
Point clean (xyzI, 512)	0.8542	0.8325
Point robust (xyzI, 512)	0.8657	0.8691

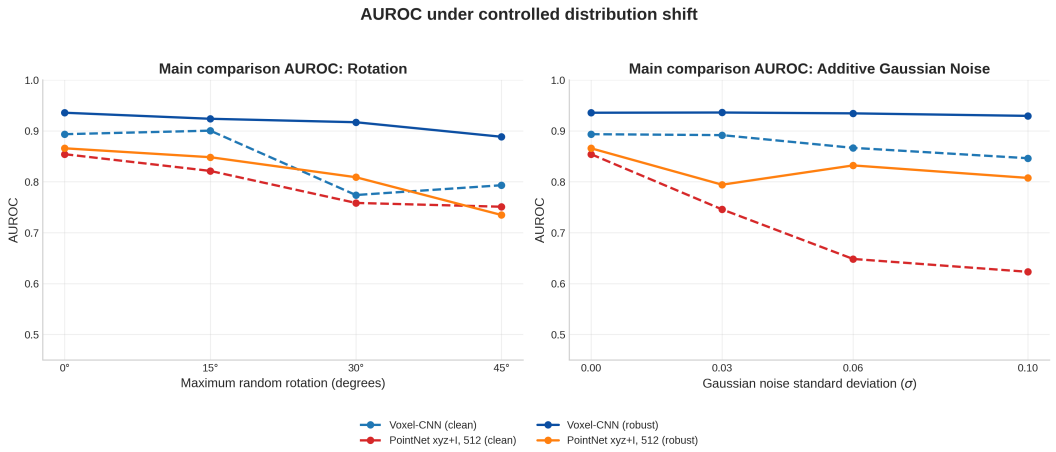

Figure 1: Main AUROC comparison under controlled shifts: rotation (left) and additive corruption (right) for voxel vs. point (xyz+I, 512), with clean vs. robust training.

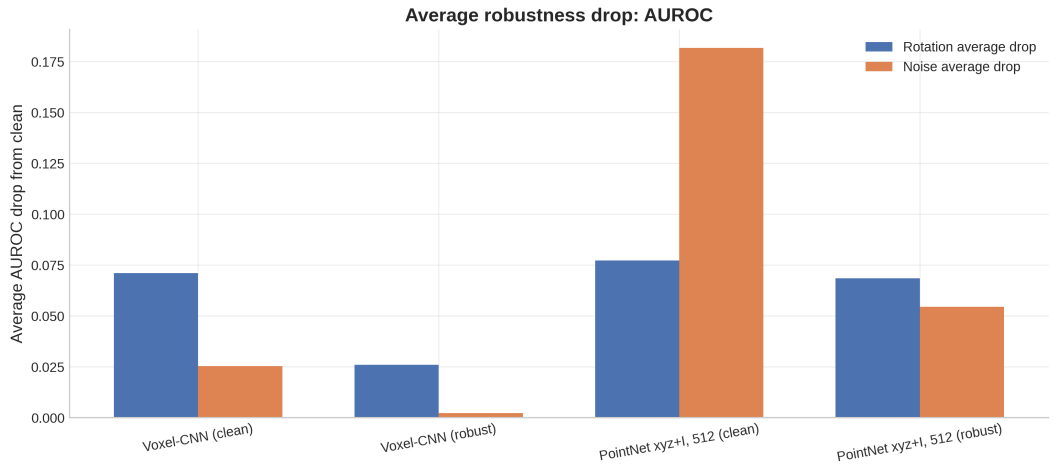

Figure 2: Average AUROC robustness drop from clean performance (lower is better), summarizing stability under rotation and additive corruption for the four main models.

Unless otherwise noted, thresholded metrics (accuracy/precision/recall/F1) use the validation-calibrated threshold, while AUROC is threshold-free.

6 Discussion

6.1 How Medical Images Are Usually Treated in Practice

Our final results are easier to interpret when compared with standard 3D medical-imaging practice. In most pipelines, scans remain in dense voxel form and are standardized through steps such as resampling, normalization, cropping or patch extraction, and augmentation before being passed to a volumetric network [3], [4]. This is the logic behind models such as 3D U-Net and nnU-Net, which keep the input as a coherent 3D volume during learning [4], [5]. Our benchmark follows a simplified version of that workflow. MedMNIST standardizes each sample to a $28 \times 28 \times 28$ volume [6]. The voxel branch stays close to that representation, while the point branch adds another conversion step by thresholding and sampling the volume into a fixed-size point cloud. This helps explain why the voxel models are more stable overall.

6.2 Why xyz+I with 512 Points Is the Best Point-Side Choice

Within the point branch, xyz+I with 512 points is still the best configuration. The key reason is that it preserves both spatial location and local intensity after voxel-to-point conversion. PointNet is designed for unordered point sets and can naturally use per-point features beyond xyz coordinates [2]. In our case, intensity retains part of the original medical-image signal that would otherwise be lost during sparsification. The advantage of 512 points over 1024 points is also plausible for this dataset. PointNet++ shows that point methods are sensitive to local structure and sampling density [7]. Because our points are extracted from already downscaled 28^3 volumes, a larger point budget does not necessarily add useful information and may instead introduce redundancy. The stronger robustness of xyz+I with 512 points therefore suggests that a smaller but more informative point set is easier for the model to use consistently.

6.3 Why the Voxel Model Still Wins Overall

Even after selecting the best point configuration, the voxel branch remains the stronger final solution. The robust voxel model reaches 0.9250 AUROC and 0.8613 accuracy on the clean test set, and with TTA it improves to 0.9588 AUROC and 0.9450 accuracy. By comparison, the xyz+I 512-point model reaches 0.8543 AUROC and 0.7225 accuracy, with a larger drop under corruption, especially additive noise. This gap is consistent with practical medical-image modeling. Dense volumetric pipelines preserve neighborhood relations, weak intensity patterns, and local 3D context that convolution can learn directly [3]–[5]. In the point pipeline, those cues must survive thresholding, sampling, normalization, and unordered aggregation before the classifier can use them. On a small standardized dataset such as MedMNIST, that extra compression is likely costly [6].

6.4 Implications for Future Point-Based Medical Models

The point results are still useful because they show that a geometry-oriented representation is viable, but they also suggest that point methods need more careful image handling than our baseline conversion. A stronger next step would be to build points from a more meaningful medical structure, such as a segmented surface, centerline, or mesh, instead of directly thresholding a small benchmark volume. Our results support that direction: xyz+I with 512 points benefits from preserving intensity and controlling sampling density, while the remaining gap to the voxel model suggests that better local neighborhood modeling is still needed. This is closely aligned with the motivation of PointNet++ [7].

Overall, the practical conclusion stays the same. For the strongest end-to-end classifier on this benchmark, the robust voxel model with TTA is the best choice because it matches how 3D medical images are most commonly prepared and modeled in practice [3]–[5]. For further point-model development, xyz+I with 512 points is the right starting point, but meaningful gains will likely require a more medically grounded point-construction pipeline.

7 Conclusions

This project evaluated voxel-based and point-based pipelines for 3D intracranial aneurysm classification on VesselMNIST3D. Both representations are viable, but they are not equally robust. The voxel branch emerged as the stronger overall solution: the robust voxel model achieved 0.9250 AUROC and 0.8613 accuracy on the clean test set, and the TTA-enhanced version improved further to 0.9588 AUROC and 0.9450 accuracy.

The point branch also improved under robust training, especially under additive noise, but remained more sensitive to feature construction, preprocessing, and threshold choice. Overall, the strongest final outcome is the robust voxel model with TTA, since it offers the best balance of predictive quality, robustness, and engineering reliability.

References

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Appendix

The result bundle contains trained checkpoints, benchmark CSV files, and generated figures. The most important supplementary artifacts are the clean summary table, the robust-vs-TTA comparison table, the point-model selection table, and the benchmark plots used in the report.

Use of AI

AI tools were used during this project as a coding, drafting, and editing assistant. AI assistance was used to help clarify implementation details, suggest code organization, improve the readability of scripts and notebook cells, address runtime issues encountered during development, and support report formatting and language refinement. AI was not used to generate experimental results. The selection of models, the design of the voxel- and point-based pipelines, the choice of benchmark settings, the interpretation of results, and all debugging, validation, and final reporting decisions were made by the group based on direct inspection of the code, figures, and exported outputs. In the writing stage, AI was used to improve the grammar, concision, and organization of the final report.

Contest for Information Sharing

As part of this course, we may share selected project materials (e.g., reports, presentation slides, and presentation recordings) on the course webpage as learning resources for future students. Additionally, we may use anonymized project information for internal statistical analysis of course outcomes. Please indicate your preferences below. *Your choices will not affect your grade in any way.*

Consent for Sharing Project Materials

All group members consent to allow the project materials (report, slides, and presentation recording) to be shared on the course page for future students.

Consent for Use of Project Information in Statistical Analysis

All group members consent to allow anonymized information about the project (e.g., topic, methods, outcomes, grades) to be used by the instructor for statistical analysis and course improvement.

Optional Comments

Please list any specific conditions or comments here.

None.

Group Identification

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